Andrii Oshlianskyi

Senior Machine Learning Engineer — Computer Vision, OCR, NLP

Kyiv, Ukraine | Remote andronex@gmail.com +380 66 920 5888 andrii-oshlianskyi

Professional Experience

Machine Learning Engineer / Researcher

Samsung R&D Institute Ukraine | *Kyiv, UA* *May 2024 – Jan 2026*

- Prepared and processed multilingual handwritten datasets; applied special handling for CJK scripts and generated synthetic data for various tasks.
- Trained OCR models for different languages and scripts using image and stroke-based (PLT) data.
- Developed language and text-type classifiers using custom architectures, including transformer-based models and various attention-based approaches.
- Built text-based autocorrection and predictive text models for OCR outputs; leveraged sequence models and LLMs.
- Conducted research on PDF reading order and document structure reconstruction using graph-based approaches (PyTorch Geometric).

Machine Learning Engineer

Preste | *Paris, FR, Remote* *Oct 2022 – May 2024*

- End-to-end development of production computer vision pipelines.
- RGB + thermal (FLIR) image preprocessing and alignment; defect detection using YOLOv8; ROI segmentation using SAM + CLIP.
- X-ray defect localization with two-stage (segmentation + classification) and one-stage UNet approaches; postprocessing using classical CV techniques.
- OCR pipelines for document processing, text extraction, and classification; backend integration.
- POC models for tiny-object and fake detection with custom preprocessing and multi-class strategies.
- Inference optimization using ONNX, quantization, and TensorRT; deployment on NVIDIA Jetson edge devices with Docker.
- Interactive demonstration and testing via Gradio applications.

Machine Learning Engineer / Data Scientist

EJaw | *Kyiv, UA* *Aug 2022 – Apr 2023*

- End-to-end development of medical and biological image classification pipelines.
- Multi-stage segmentation and classification pipelines for cervical and blood cell analysis; sequence modeling using CRNN, BLSTM, and Attention mechanisms.
- Model explainability and validation using Grad-CAM and saliency maps.
- R&D on drone-based 3D reconstruction, object recognition, and autonomous navigation.
- Generative modeling for design using Stable Diffusion + ControlNet.

Machine Learning Engineer

DeepX | *London, UK, Remote* — *Oct 2021 – Jul 2022*

- Object detection and tracking pipelines for airport security video analysis using YOLOv5 and classical CV tracking techniques.
- Multi-camera video processing pipelines for automated vending machine applications; handled synchronization, low-FPS frame analysis, and distortion correction.
- Pose estimation and event recognition combining deep learning models and classical computer vision algorithms.
- Full pipeline development with decision logic, model optimization, and deployment-ready implementations.

Research Internships

- **GSI Helmholtz Centre for Heavy Ion Research** | *Darmstadt, DE* — *Jul 2018 – Sep 2018*
  Monte Carlo simulations and data analysis.
- **Computational Physics Department, University of Oslo** | *Oslo, NO* — *Feb 2018 – Apr 2018*
  Numerical simulations in theoretical physics models.

Education, Research & Publications

Master of Science in Physics and Astronomy

Taras Shevchenko National University of Kyiv | *Kyiv, UA* — *2019 – 2021*
Nuclear Physics, High Energy and Particle Physics
- Diploma with Honour

Bachelor of Science in Physics

Taras Shevchenko National University of Kyiv | *Kyiv, UA* — *2015 – 2019*
Nuclear Physics, High Energy and Particle Physics
- Diploma with Honour

During my studies, I participated in multiple international conferences, workshops and scientific schools and co-authored 2 publications in peer-reviewed journals and contributed to several conference proceedings, gaining experience in high-energy and particle physics research.

Publications

- Astrophysical Journal, Vol. 916 (2021), Article 15. Supernova Model Discrimination with Hyper-Kamiokande.
- Proceedings of Science, Vol. 395 (2021), Article 1193. Prospects for Neutrino Astrophysics with Hyper-Kamiokande. 37th International Cosmic Ray Conference (ICRC 2021).

Skills

Languages: English (C1), Ukrainian (native), Russian (fluent)

Programming Languages: Python, C++, Bash

Frameworks and Libraries: PyTorch, Lightning, Transformers, Hydra, Optuna, TensorFlow, Keras, YOLO, SAM, CLIP, GradCAM, NLTK, spaCy, tokenizers, BERT, SentenceTransformers, OpenCV, Albumentations, NumPy, Pandas, SciPy, scikit-image, matplotlib, seaborn, TensorBoard, TorchMetrics, PyTorch Geometric, WandB, MLflow, Poetry

MLOps and Deployment Docker, Singularity, DVC, ONNX, TensorRT, NVIDIA Jetson, JetPack SDK; Gradio, Flask(base), AWS(base), Azure (base)